

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Computer Science)	Semester:	Spring 2026
	Duration:	15 minutes	Total Marks:	15
	Paper Date:		Section	6B1
	Exam Type:	Quiz 6 - Chapter 6	Page(s):	2

Student Name

Roll No.

Section:

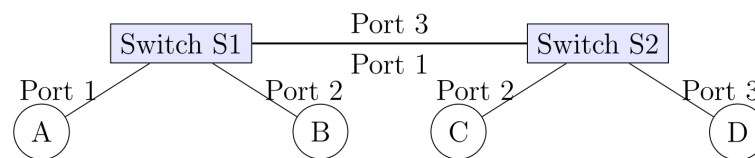
Q1. Carefully read the Question before attempting:

[5 marks]

Consider the switched LAN topology shown below. The switch tables are initially empty. The following sequence of frame transmissions occurs:

1. Node A sends a frame to Node D.
2. Node D replies with a frame to Node A.
3. Node C sends a frame to Node D.

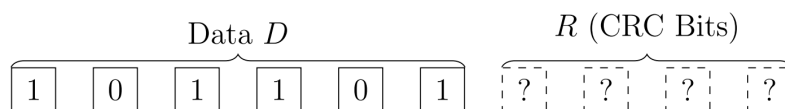
Show the state of the Switching Table for **Switch S2** after these three events. Format: (MAC Address, Interface). Also, show the same for **Switch S1**.



Q2. Carefully read the Question before attempting:

[10 marks]

Consider a data transmission scenario using a CRC generator polynomial $G(x) = x^4 + x + 1$. The data string to be transmitted is $D = 101101$.



1. Analytically calculate the 4-bit CRC remainder R . Show the long division in binary modulo-2 arithmetic. What is the actual bit-string transmitted by the sender?
2. A "burst error" of length k is a contiguous sequence of bits in which the first and last bits are errors, and the intermediate bits may or may not be errors. Can this specific generator $G(x)$ can detect **all** burst errors of length $k = 3$. Give a reason?
3. Suppose a burst error occurs during transmission such that the received bit string has the 3rd and 4th bits (counting from the left, 1-based index) inverted. Does the receiver accept or reject this frame?